# The empirical recurrence for OEIS A221796, proved by transfer matrix

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3 September 2026

**Abstract**

OEIS A221796 counts the distinct OCCUPANCY arrays obtained when every cell of an $n \times 3$ grid holds one token, each token moves to a neighbouring cell, and the direction a token leaves a cell in is restricted by the direction in which another token arrives there. The entry carries an empirical recurrence of order 10 contributed by R. H. Hardin. It is true, and it is decidable rather than empirical. What is counted is the size of an IMAGE — two different assignments of moves with the same occupancy count once — so it is not a walk count in the obvious graph; determinising the machine that emits the occupancy row by row turns it into one, on $S = 163$ states, after which the conjectured recurrence is settled by a finite exact computation.

2020 Mathematics Subject Classification. 05A15, 68Q45, 60C05.

## 1 The conjecture

OEIS A221796 is "Number of n X 3 arrays of occupancy after each element moves to some horizontal or vertical neighbor, without move-in move-out straight through or left turns.". It has offset 1 and begins

$$0, 12, 12, 207, 470, 4073, 13314, 85974, \ \ldots$$

The entry states, as a conjecture contributed by its author and never marked settled:

> Empirical: a(n) = 27*a(n-2) + 14*a(n-3) - 165*a(n-4) - 20*a(n-5) + 329*a(n-6) - 48*a(n-7) - 131*a(n-8) - 34*a(n-9) + 9*a(n-10).

As of the "Last modified" line on the live entry (Aug 11 2018 07:13:51, revision 8) this is still recorded as empirical, and nothing on the entry records it as proved.

## 2 What is being counted

Every cell of an $n \times 3$ grid holds one token. Each token moves by one of the offsets

$$(-1, 0), (0, -1), (0, 1), (1, 0),$$

a move being available only when it lands inside the grid. The offset $(0, 0)$ is NOT among them: every token must move. The OCCUPANCY array records, for each cell, how many tokens land on it, and the entry counts how many different occupancy arrays occur as the assignment of moves runs over all its possibilities.

The moves are not free of one another. Suppose the token at a cell $A$ moves by $d$ to the cell $B$, and the token standing at $B$ moves by $d'$. The entry forbids the ordered pair $(d, d')$ when

- $d \wedge d' > 0$, that is, the outgoing direction lies to the LEFT of the incoming one, the wedge being the cross product of the two offsets read as plane vectors;
- $d' = d$, that is, the token arriving at a cell and the token leaving it move in the same direction — the move goes STRAIGHT THROUGH;

and imposes nothing else. A token that stays put neither arrives anywhere nor leaves, so a zero offset on either side of the pair carries no condition at all.

The sense of "left" is fixed once and for all. Rows are numbered downwards, so the offset $(d_1, d_2)$ is the plane vector $(d_2, -d_1)$, and $d \wedge d'$ denotes $d_2(-d'_1) - (-d_1)d'_2$. A positive value says $d'$ is obtained from $d$ by a rotation through an angle strictly between 0 and $\pi$ counterclockwise, which is what a turn to the left is. The condition is not a 90° test: for a neighbour set containing diagonal offsets the turns of 45° and 135° are left turns too, and the entry's data distinguishes the two readings.

That the count is the size of an IMAGE is the whole difficulty: two assignments with the same occupancy must be counted once, and a walk count over the assignments would count them twice.

# 3 One window decides everything

Write $c_i \in \Delta^3$ for the row of offsets chosen by the cells of row $i$, where $\Delta$ is the offset set above.

**Lemma 1.** *Both the occupancy of row $i$ and every instance of the forbidden pair whose FIRST member is a move made in row $i$ are determined by $c_{i-1}, c_i, c_{i+1}$, with the convention that an absent row is the symbol $\perp$.*

*Proof.* Every offset changes the row index by at most one, so a token landing on a cell of row $i$ started in row $i-1$, $i$ or $i+1$; that is the occupancy. For the pairs, a move made by the cell $(i, j)$ with offset $d$ lands on the cell $(i + d_1, j + d_2)$, whose row is $i-1$, $i$ or $i+1$, so the offset $d'$ of that cell is a coordinate of $c_{i-1}$, $c_i$ or $c_{i+1}$. Reading a row of the array as $\perp$ makes "up" or "down" unavailable, which is exactly the boundary rule. □

Each ordered pair is therefore tested exactly once, from the row of the token that makes the first move of the pair; every row plays that part exactly once along the array.

Let $\beta(p, c, x)$ be the occupancy row of $c$ in the context $(p, x)$, undefined when some cell of $c$ has no room to move or when some forbidden pair occurs. The occupancy array of an assignment $c_1, \ldots, c_n$ is

$$\big(\beta(\perp, c_1, c_2),\ \beta(c_1, c_2, c_3),\ \ldots,\ \beta(c_{n-1}, c_n, \perp)\big).$$

# 4 The image is a walk count after determinisation

Determinise. For a set $D$ of pairs $(p, c)$ and an occupancy row $b$ put

$$\delta(D, b) = \{(c, x) : (p, c) \in D,\ \beta(p, c, x) = b\}, \qquad f(D) = \big|\{\beta(p, c, \perp) : (p, c) \in D,\ \beta(p, c, \perp) \text{ defined}\}\big|,$$

and let $D_0 = \{(\perp, c) : c\}$. An induction on the number of rows shows $D_t$ is exactly the set of pairs of consecutive move-rows completing a prefix with the emitted occupancy rows, so that with $M$ the adjacency matrix of the digraph on the reachable nonempty sets, one edge per occupancy row it admits,

$$a(n) \;=\; \iota^\top M^{n-1} \tau, \qquad \iota = e_{D_0}, \quad \tau = f.$$

The last row is counted by $f$ and not by an edge: it is emitted with nothing below it, so it is a property of the state the walk stops in and $\tau$ is not the all-ones vector. There are $S = 163$ reachable sets. In particular $a$ is $C$-finite.

## 5 The proof

**Lemma 2.** *Let $q(t) = t^{10} - \sum_i c_i t^{10-i}$ be the characteristic polynomial of the conjectured recurrence. The residual $a(n) - \sum_i c_i a(n-i)$ equals $\iota^\top M^j q(M)\tau$ for the corresponding walk index $j$, so the recurrence holds from some point on if and only if $\iota^\top M^j q(M)\tau = 0$ for all large $j$, and it is enough to examine $j < S$.*

*Proof.* Factor $M^j$ out of $M^{j+10} - \sum_i c_i M^{j+10-i}$. For the bound, $M^S$ is an integer combination of $I, M, \ldots, M^{S-1}$ by Cayley–Hamilton, so the numbers $u_j = \iota^\top M^j q(M)\tau$ obey the monic recurrence given by the characteristic polynomial of $M$; $S$ consecutive zeros force every later one, and the last nonzero $u_j$ pins the threshold exactly. □

**Theorem 3.**

$$a(n) \;=\; 27\,a(n-2)+14\,a(n-3)-165\,a(n-4)-20\,a(n-5)+329\,a(n-6)-48\,a(n-7)-131\,a(n-8)-34\,a(n-9)+9\,a($$

*for every $n > 10$.*

*Proof.* The vector $w = q(M)\tau$ was formed by 10 matrix–vector products in exact integer arithmetic and $\iota^\top M^j w$ evaluated for $j = 0, 1, \ldots,$ again exactly; those integers vanish from the index corresponding to $n = 10 + 1$ onwards and the run continued until the working vector was identically zero, which settles every larger $j$ at once. □

## 6 Verification

Four checks, each independent of the algebra above.

First, the model reproduces all 22 terms the entry publishes, exactly, in integer arithmetic.

Second, the count was recomputed for the smallest grids straight from the definition — enumerate every assignment of offsets to cells, discard the assignments containing a forbidden pair, form the occupancy array, and count the distinct arrays — with the forbidden-pair test written out from the entry's wording rather than taken from the model. The published terms came back.

Third, the conjectured recurrence was evaluated directly on the published terms, with no matrices involved, and holds wherever the proved range and the data overlap.

Fourth, the annihilation test of Lemma 2 was carried out in exact integer arithmetic on the whole vector rather than on a sampled prefix.

## References


[1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, https://oeis.org, 2026. Sequence A221796.

[2] J. E. Hopcroft, R. Motwani and J. D. Ullman, *Introduction to Automata Theory, Languages, and Computation*, 3rd ed., Addison–Wesley, 2006. (The subset construction, Section 2.3.)

[3] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)

[4] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.